



BEST PRACTICE SERIES

Saving Hydraulic Oil during
Inspect Torque Converter
Sump Screen

Maintenance and Repair Application Component Life Management Component Renewal (CRC) MARC Management

Saving Hydraulic Oil during Inspect Torque Converter
Sump Screen0

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1.0 Introduction

Inspecting the Torque Converter Sump Screen is one of the Preventive Maintenance tasks and should be a part of the Condition Monitoring program. Dealer experience indicates to remove the screen for inspection, it is necessary to drain approximately 10 L (2.6 Gal) of oil from the torque converter housing. This oil is then discarded and new oil is used to adjust the oil level.

2.0 Best Practice Description

This Best Practice outlines the parts and procedure to drain the oil and pump into the tank utilizing a kidney loop tool. Using this procedure minimizes the risk of spilling oil and eliminates wasted oil.

3.0 Implementation Steps

To implement this Best Practice:

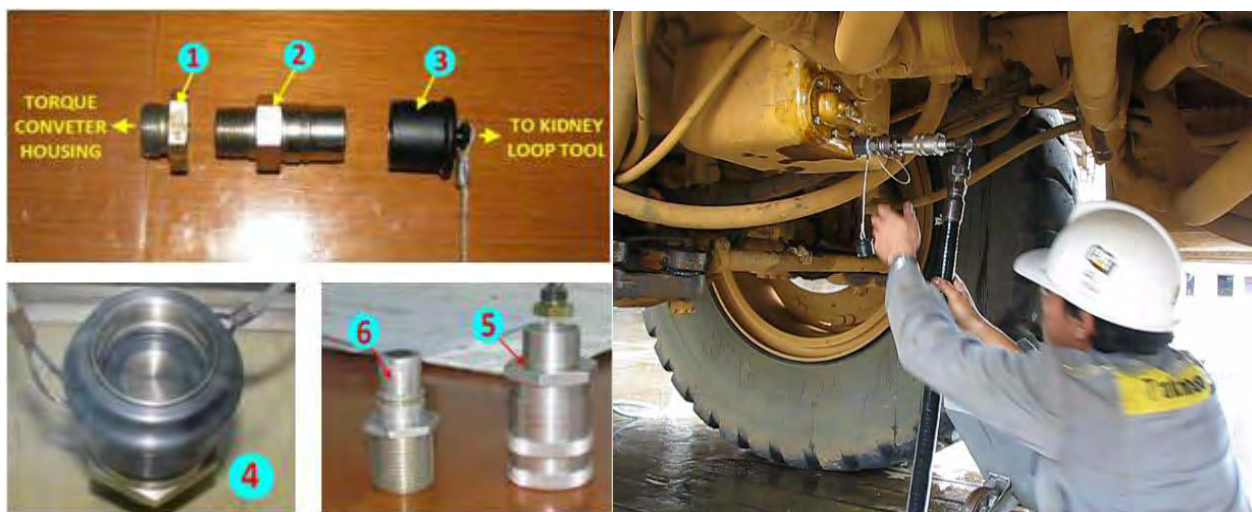
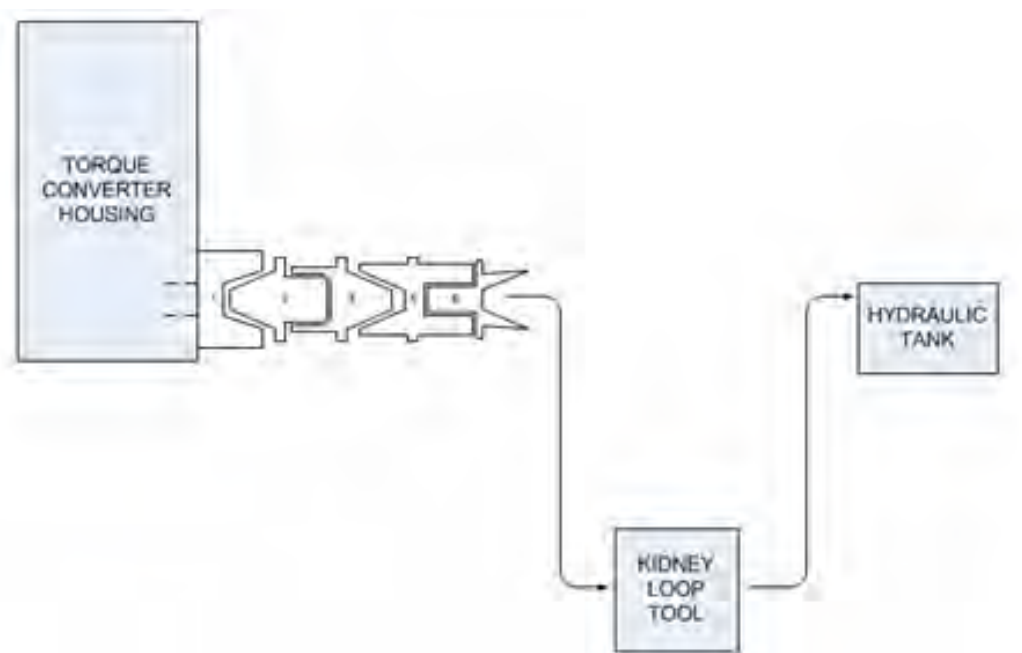
1. This method can be performed on the following trucks:

MODEL	PREFIX
777B	4YC
777C	4XJ
777D	2YW, 3PR, AGC, AGY, FKR
777F	JRP
785	8GB
785B	6HK
785C	1HW, 5AZ, APX
785D	DMC, MSY
789	9ZC
789B	7EK
789C	2BW
793	3SJ
793B	1HL

2. Install the appropriate adapter and connectors on each truck in the fleet. These parts are available from Caterpillar. Reference part numbers listed below.
3. Utilize a kidney loop tool to drain oil from the torque converter housing and pump into the hydraulic tank.
4. Remove the screen for inspection and clean if necessary and then reinstall.
5. Check oil levels and adjust if necessary.

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4.0 Benefits

Although additional parts must be purchased and installed, there is no additional labor time needed during the PM to perform this procedure. No oil is discarded so there is a savings in the amount of oil needed to perform the screen inspection during PM. Since a connector is used instead of the drain plug, you no longer have to replace the drain plug gasket at each PM.

Example for a fleet of 50 X 777C/D:

Assume the screen is inspected at each PM and the PM interval is 330 hours. Approximately 10 liters of oil is discarded at each PM. This results in 884 PM events per year for the fleet and 8,840 liters of discarded oil.

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The tables below show the potential annual savings for this fleet utilizing information provided by Trakindo. The currency has been converted to US dollars.

Description	Oil Qty / PM event	PM Events (fleet)	Total Oil Drained	Oil cost per Liter (US \$)	Total Oil Cost (US \$)
Hydraulic oil drained when checking T/C screen	10 L	884	8,840 L	\$4.40	\$38,896

Using the drain plug, the 2A-3398 gasket should be replaced each PM. Utilizing this Best Practice, the drain plug is not used so there is no gasket to replace which increases the savings.

Part Number	Description	Qty	Price (US \$)	Total (US \$)
2A-3398	Gasket	884	\$3.20	\$2,828.80

NOTE: Benefits will vary depending upon inspection frequency of the torque converter sump screen and whether inspection is performed during torque converter oil change.

5.0 Resources Required

This simple installation requires 4 parts, i.e.:

1. Bushing
2. Coupler - male
3. Cap As
4. Coupler – female (for kidney loop tool)

No.	Part Number	Description	Qty
1	3R-7117	Bushing	1 per truck
2	9S-9271	Coupler - male	1 per truck
3	9S-9272	Cap as	1 per truck
4	9S-9273	Coupler - female	1
5	126-7538	Nozzle as	1
6	201-8057	Cap	1

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6.0 Supporting Attachments / References

None

7.0 Related Best Practices

None

8.0 Acknowledgements

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